

Assignment 7

Replicating Interaction Techniques (25P)

Hand in groups of two. Do not make teams on your own!

Goals

You understand how complex interaction techniques can be implemented. You can implement interaction techniques based on their description in a scientific paper and supplementary material.

1. Paper Selection (4P)

Search for scientific papers that describe the concept and/or implementation of new interaction techniques. **Demonstration** and **short-paper** tracks at conferences such as UIST, EICS, MobileHCI, and CHI might be a good starting point. You can also use projects discussed in our journal club or replicated in previous ITT courses. We will provide you with a list of previous replications in Ilias.

Compare several candidates you find interesting. Then, exclude everything that cannot be implemented within **two weeks**. This can be due to hardware requirements or to the need to gather a lot of training data first. If you are unsure whether the scope of your selected paper is too long or too short, you can write to us. However, feel free to build upon existing code and/or training data. Then, decide which prototype you want to implement and/or extend. **Important: When you selected a paper enter it in the “Ausgewählte Paper”-Tabel in ILIAS.** Every Paper should only be replicated once so check which paper other teams replicate. First come first serve.

Document this selection process in your **documentation.md**. Also, create an entry in the Ilias data table for the paper you will replicate.

Score

- (1P)** At least two suitable papers were considered (and shortly discussed in the documentation)
- (2P)** The final paper selection is well justified (strength, scope is neither too long nor too short, ...).
- (1P)** Sources are cited for the selected paper and the alternatives considered

2. Implementation (10P)

Implement or extend the interaction technique described in your selected paper. You do not have to build a feature-complete copy, but the general interaction should be similar. If you decide to extend an existing project, your addition should be focused on interaction. All source code required to run the project must be included in your repository. If you want to make fundamental changes, discuss them with us beforehand.

Score

- (2P)** The project runs successfully without debugging
- (7P)** The core interaction technique is implemented correctly relative to the chosen paper
- (1P)** The code is well structured, readable and organized

3. Documentation (5P)

Document your complete development process in the **documentation.md**. The documentation should not only explain the final implementation but also decisions you made throughout the project. Write it so that every other student can understand your design decisions and reproduce your work based on your documentation.

Furthermore, reflect on the challenges you encountered, how you solved them and what limitations remain in your prototype.

Score

- (1P)** The development process and design decisions are explained and understandable
- (3P)** The documentation is extensive: it shows not just the final product but also iterations, intermediate states, possible problems encountered and solutions, ...
- (1P)** Setup, dependencies and steps are clear enough that any other person could rebuild the work

4. Presentation (6P)

Present your work at the last ITT session **on July 21st**. So, each group must be present during this session.

In your presentation, you should briefly introduce the chosen paper and explain your implementation process. The main focus should be a **live demonstration** of your prototype. We want to see the interaction technique in action.

The presentation should last **8 - 10 minutes**.

After the presentation, there will be a short demo session in which everyone will test the other prototypes.

Score

- (1P)** Appropriate length of presentation
- (1P)** Good presentation overall
- (4P)** Live-Demo works and shows the interaction technique